

Intro to MapReduce

Sure! Let's talk about MapReduce in simple terms.

MapReduce is a way to process large amounts of data by breaking it down into smaller, manageable pieces. Imagine you have a big pile of Lego blocks, and you want to count how many of each color you have. Instead of counting them all at once, you can split the pile into smaller groups. This is like the "Map" step, where each group of blocks is processed to create a list of colors and their counts. For example, you might end up with a list that says "Red: 5, Blue: 3, Green: 2."

Once you have all these smaller lists, the "Reduce" step comes in. This is where you take all the lists from the Map step and combine them to get a final count. So, if you had three groups that counted Red blocks as 2, 3, and 5, the Reducer would add those together to give you a total of 10 Red blocks. This way, MapReduce helps you efficiently process and analyze large datasets by breaking them down and then combining the results.

What is MapReduce?



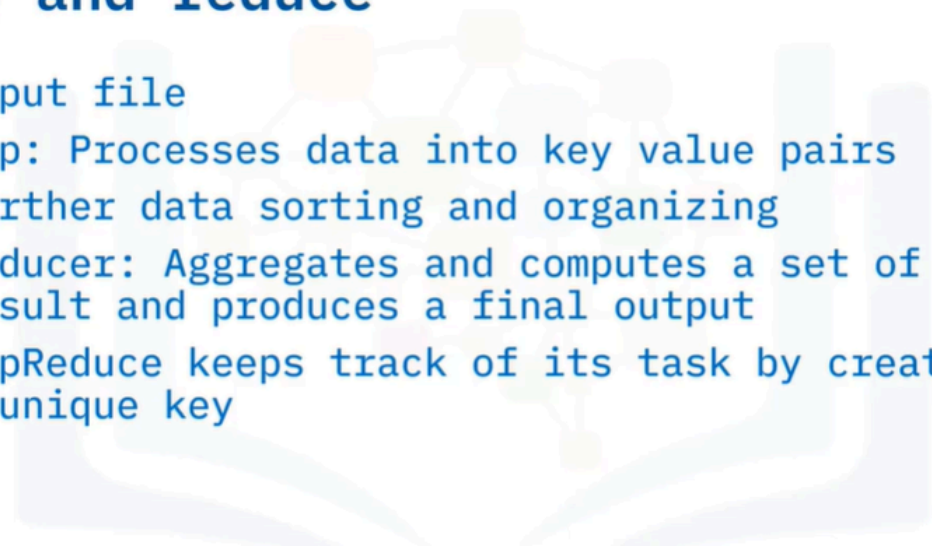
Distributed computing is a system with multiple components located on different machines that communicate actions in one view to the end user

- Programming model used in Hadoop for processing Big Data
- Processing technique for distributed computing
- Consists of a Map task and a Reduce task
- Can be coded in many programming languages like Java, C++, Python, Ruby and R

- What is MapReduce? MapReduce is a programming pattern that enables massive scalability across hundreds or thousands of servers in a Hadoop

cluster. As the processing component, MapReduce is the heart of Apache Hadoop. MapReduce is a processing technique and a program model for distributed computing, it is based on Java. Distributed computing is a system or machine with multiple components located on different machines. Each component has its own job, but the components communicate with each other to run as one system to the end user." The MapReduce algorithm consists of two important tasks - Map and Reduce. Many MapReduce programs are written in Java. MapReduce can also be coded in C++, Python, Ruby, R and so on.

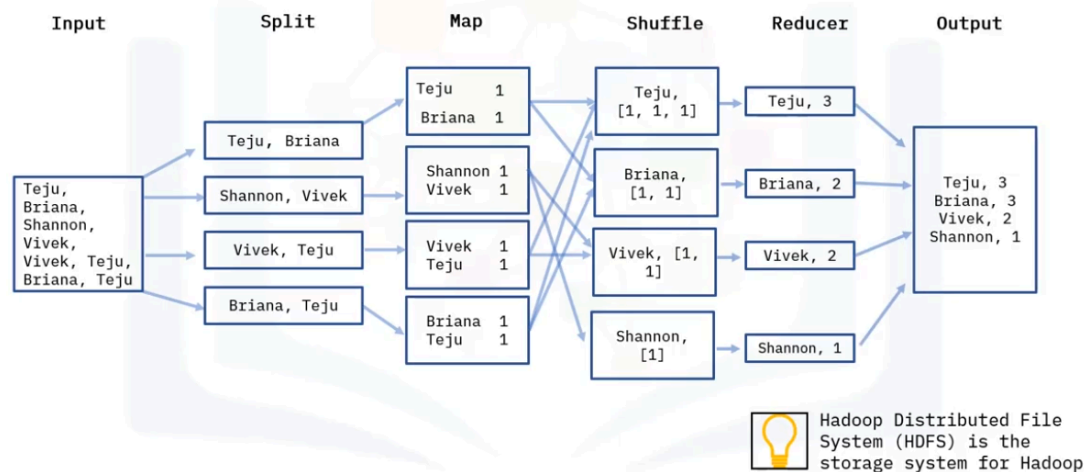
Map and reduce

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- Input file
 - Map: Processes data into key value pairs
 - Further data sorting and organizing
 - Reducer: Aggregates and computes a set of result and produces a final output
 - MapReduce keeps track of its task by creating a unique key

- As the name suggests, the MapReduce framework contains two tasks, Map and Reduce.
- Map takes in an input file and performs some mapping tasks by processing and extracting important data information into a key value pairs and these are the preliminary output list. Some more reorganization goes on before the preliminary output is sent to the Reducer.
- The Reducer works with multiple map functions and aggregates the pairs using their keys to produce a final output. MapReduce keeps track of its tasks

by creating unique keys to ensure that all the processes are solving the same problem.

How MapReduce works



- Let us look at the framework visually. First, we have the Map step, which takes a set of data and converts it into another set of data, where individual elements are broken down into key/value pairs. The key is the name, and the value is the content. The input data is a file that is saved in the Hadoop file system called HDFS.
- Now let's assume we have an input file that contains names of people, and we would like to do a word count on the unique name occurrences. First, the data is split into the following four files, each of them in key value pair and are worked on separately. For example, for the first split line for Teju and Briana, we have two key value pairs with one occurrence in each file, and it will do the same for all of key value pairs.
- We then have the reducer; the reducer processes the data that comes from the map. After processing, it produces a new set of outputs, which will be stored in the HDFS. The Reducer starts with shuffling. Shuffling sorts the key and a list of values in a list, for example, you will see the Key Teju and the corresponding list of values from the previous step. We will have Teju [1, 1, 1], This is because the name Teju occurred 3 times in the "Map" step. It does the

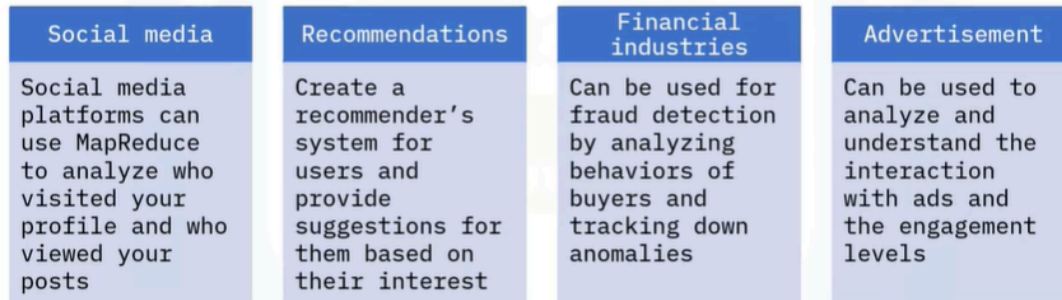
same for the rest of the names, counting how many times they appeared in the “Map” step. The Reducer layer then aggregates the values in the list and saves them, then the final output is saved in an output file.

Why use MapReduce?



- The advantages of MapReduce is its ability to allow for a high level of parallel jobs across multiple nodes. A node is an independent computer used for processing and storing big volumes of data.
- In Hadoop we have two types of nodes, the name node and the data node. Map reduce allows for splitting and running independent tasks in parallel by dividing each task which in turn saves time. MapReduce is very flexible and can process data that come in tabular and non-tabular forms. Therefore, MapReduce provides business value to organizations regardless of how their data is structured. It also offers support for different languages and provides a platform for analysis, data warehousing and more.

Common use cases



Social media	Recommendations	Financial industries	Advertisement
Social media platforms can use MapReduce to analyze who visited your profile and who viewed your posts	Create a recommender's system for users and provide suggestions for them based on their interest	Can be used for fraud detection by analyzing behaviors of buyers and tracking down anomalies	Can be used to analyze and understand the interaction with ads and the engagement levels

- MapReduce has a couple of use cases and here we have some of them displayed. MapReduce can be used for social media platforms like LinkedIn and Instagram to analyze who visited, viewed, and interacted with your profile posts. Map reduce is used by Netflix to recommend movies based on what you have watched in the past by using the user's interests. It is also used in financial institutions like banks and credit card companies to flag and detect anomalies in user transactions. It can also be used in the advertisement industry to understand a user's behavior by how they engage with ads.

Summary

In this video, you learned that:

- MapReduce is a framework used in parallel computing
- MapReduce contains two major tasks, map and reduce
- MapReduce is flexible for all data types
- Can be applied to multiple industries such as social media, entertainment, and many more